

Lab 2: Website Enumeration and Information Gathering

Tools Used

- Kali Linux
- Whatweb

Exercise 1:

whatweb command to detect technologies for the following targets:

- example.com
- stackoverflow.com
- github.com

Command Used: whatweb example.com, whatweb stackoverflow.com, whatweb github.com

```

(kali@kali)-[~]
$ whatweb example.com
http://example.com [200 OK] Allow[GET, HEAD], Country[RESERVED][22], HTML5, HTTPServer[cloudflare], IP[172.66.147.243], Title[Example Domain], UncommonHeaders[cf-cache-status,cf-ray]
https://example.com [200 OK] Allow[GET, HEAD], Country[UNITED STATES][US], HTML5, HTTPServer[cloudflare], IP[104.20.23.154], Title[Example Domain], UncommonHeaders[cf-cache-status,cf-ray]

(kali@kali)-[~]
$ whatweb stackoverflow.com
http://stackoverflow.com [301 Moved Permanently] Cookies[_cf_bm], Country[UNITED STATES][US], HTTPServer[cloudflare], HttpOnly[_cf_bm], IP[198.252.206.1], RedirectLocation[https://stackoverflow.com/], Title[301 Moved Permanently], UncommonHeaders[x-dns-prefetch-control,cf-ray]
https://stackoverflow.com [403 Forbidden] Cookies[_cf_bm], Country[UNITED STATES][US], HTML5, HTTPServer[cloudflare], HttpOnly[_cf_bm], IP[198.252.206.1], Script, Title[Just a moment ...], UncommonHeaders[accept-ch,cf-mitigated,content-security-policy,critical-ch,cross-origin-embedder-policy,cross-origin-opener-policy,cross-origin-resource-policy,origin-agent-cluster,permissions-policy,referrer-policy,server-timing,x-content-type-options,x-dns-prefetch-control,cf-ray], X-Frame-Options[SAMEORIGIN], X-UA-Compatible[IE=Edge]
https://stackoverflow.com/ [403 Forbidden] Country[UNITED STATES][US], HTML5, HTTPServer[cloudflare], IP[198.252.206.1], Script, Title[Just a moment ...], UncommonHeaders[accept-ch,cf-mitigated,content-security-policy,critical-ch,cross-origin-embedder-policy,cross-origin-opener-policy,cross-origin-resource-policy,origin-agent-cluster,permissions-policy,referrer-policy,server-timing,x-content-type-options,x-dns-prefetch-control,cf-ray], X-Frame-Options[SAMEORIGIN], X-UA-Compatible[IE=Edge]

(kali@kali)-[~]
$ whatweb github.com
http://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[140.82.121.4], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[140.82.121.4], Open-Graph-Protocol[object][1401488693436528], OpenSearch[opensearch.xml], Script[application/javascript,application/json], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[140.82.121.4], Open-Graph-Protocol[object][1401488693436528], OpenSearch[opensearch.xml], Script[application/javascript,application/json], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]

```

Record of Findings:

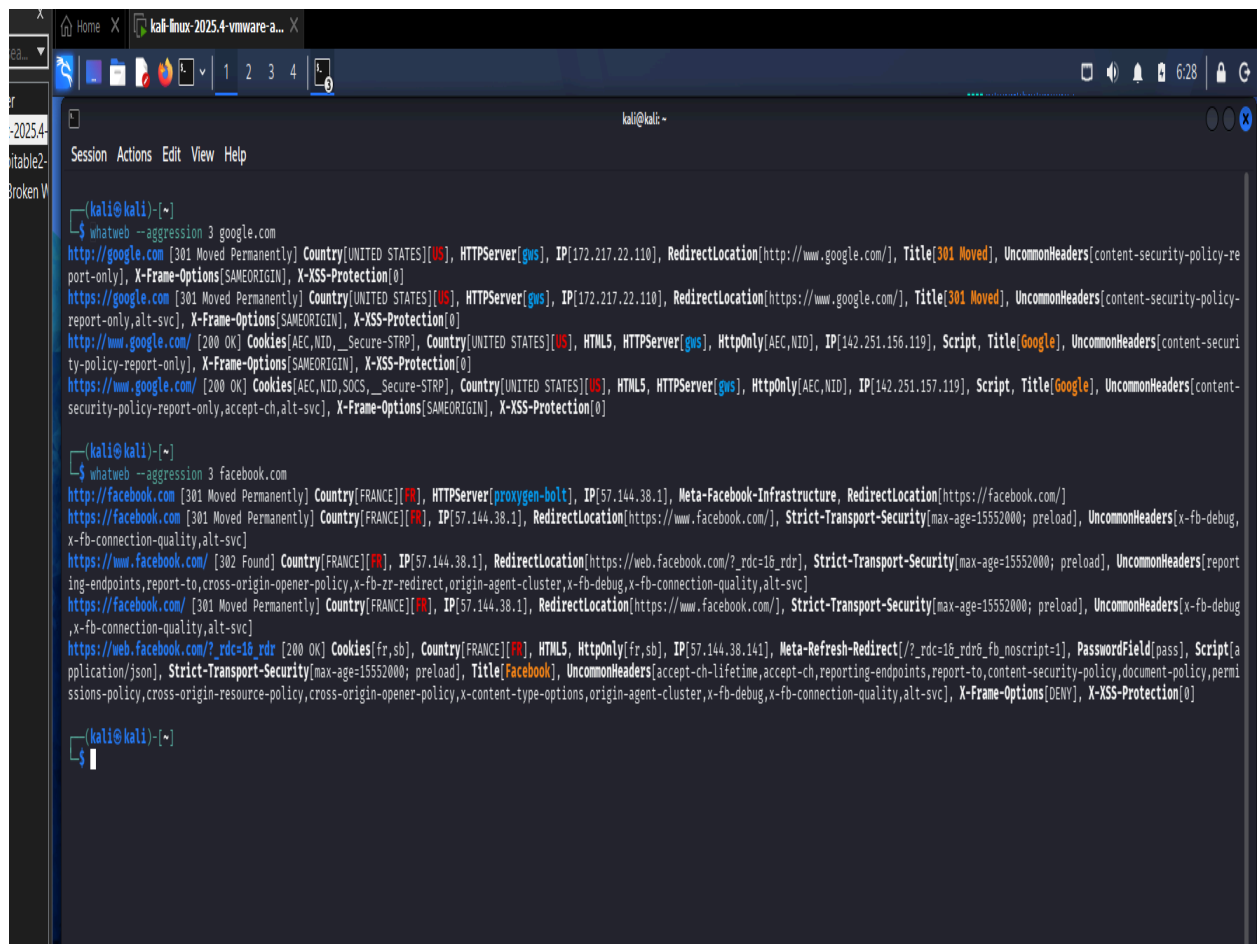
1. example.com: HTML5 and Cloudflare
2. stackoverflow.com: HTML5,Cloudflare,Script
3. github.com: HTML5,JavaScript, JSON,Open Graph Protocol, OpenSearch

Exercise 2:

Aggressive scan on the following targets:

- google.com
- facebook.com

Commands Used: whatweb --aggression 3 [google.com](https://www.google.com/), whatweb --aggression 3 [facebook.com](https://www.facebook.com/).



```
(kali@kali)~$ whatweb --aggression 3 google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[172.217.22.110], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[172.217.22.110], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID,Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.156.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://www.google.com/ [200 OK] Cookies[AEC,NID,SOCS,Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.157.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]

(kali@kali)~$ whatweb --aggression 3 facebook.com
http://facebook.com [301 Moved Permanently] Country[FRANCE][FR], HTTPServer[proxygen-bolt], IP[57.144.38.1], Meta-Facebook-Infrastructure, RedirectLocation[https://facebook.com/]
https://facebook.com [301 Moved Permanently] Country[FRANCE][FR], IP[57.144.38.1], RedirectLocation[https://www.facebook.com/], Strict-Transport-Security[max-age=15552000; preload], UncommonHeaders[x-fb-debug,x-fb-connection-quality,alt-svc]
https://www.facebook.com/ [302 Found] Country[FRANCE][FR], IP[57.144.38.1], RedirectLocation[https://web.facebook.com/?_rdc=1b_rdr], Strict-Transport-Security[max-age=15552000; preload], UncommonHeaders[reporting-endpoints,report-to,cross-origin-opener-policy,x-fb-zr-redirect,origin-agent-cluster,x-fb-debug,x-fb-connection-quality,alt-svc]
https://facebook.com/?_rdc=1b_rdr [200 OK] Cookies[fr,sb], Country[FRANCE][FR], HTML5, HttpOnly[fr,sb], IP[57.144.38.141], Meta-Refresh-Redirect[?_rdc=1b_rdr&fb_noscript=1], PasswordField[pass], Script[application/json], Strict-Transport-Security[max-age=15552000; preload], Title[Facebook], UncommonHeaders[accept-ch,reporting-endpoints,report-to,content-security-policy,document-policy,permissions-policy,cross-origin-resource-policy,cross-origin-opener-policy,x-content-type-options,origin-agent-cluster,x-fb-debug,x-fb-connection-quality,alt-svc], X-Frame-Options[DENY], X-XSS-Protection[0]
```

Findings:

1. google.com: HTML5 and Script
2. facebook.com: HTML5, Script, JSON, Meta-Facebook-Infrastructure, Meta-Refresh-Redirect

Lab 3: Subdomain Hunting

Tools Used

- Kali Linux
- sublist3r
- dirb
- KnockPy

Exercise 1:

sublist3r command for the following domains:

- github.com
- google.com

Findings:

1. Subdomains for github.com:
135 unique subdomains were discovered.
2. Subdomains for google.com:
180 unique subdomains were discovered.

Exercise 2:

Directory discovery scan on the following targets:

Commands used:

- dirb http://example.com`
- dirb <http://example.org>`

```

kali@kali: [~]
(kali@kali)-[~]
$ dirb https://example.com

DIRB v2.22 example.com
By The Dark Raver recon
3.0
Threads 250

START_TIME: Fri May 22 07:39:19 2026
URL_BASE: https://example.com/ (Windows NT 10.0; Win64; x64) AppleWebKit
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt /usr/share/dirb/wordlist/word
Reports DB: ~/knockpy/reports.db
False

GENERATED WORDS: 4612

Scanning URL: https://example.com/

Mode Status
END_TIME: Fri May 22 07:58:02 2026
DOWNLOADED: 4612 - FOUND: 0
(kali@kali)-[~]
$

Scanning domains 1/1 00000
Domain IP

```

```
Session Actions Edit View Help
(kaliⓈkali)-[~] le.com recon
$ dirb http://example.org

DIRB v2.22
By The Dark Raver example.com
recon
Timeout: 3.0
START_TIME: Fri May 22 07:52:45 2026
URL_BASE: http://example.org/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt ; x64) AppleWeb
Wordlist: /usr/lib/python3/dist-packages/knockpy/wordlist/wo
~/.knockpy/reports.db
Silent: False
GENERATED WORDS: 4612

Scanning URL: http://example.org/

END_TIME: Fri May 22 08:09:17 2026
DOWNLOADED: 4612 - FOUND: 0

(kaliⓈkali)-[~] disabled
$ disabled
disabled
```

Provider	Status
VirusTotal	X disabled
Shodan	X disabled

Findings:

1. Directories for example.com:
 - dirb tested 4612 possible directories/files but found nothing on <https://example.com>
2. Directories for example.org
 - dirb tested 4612 possible directories/files but found nothing on <https://example.com>

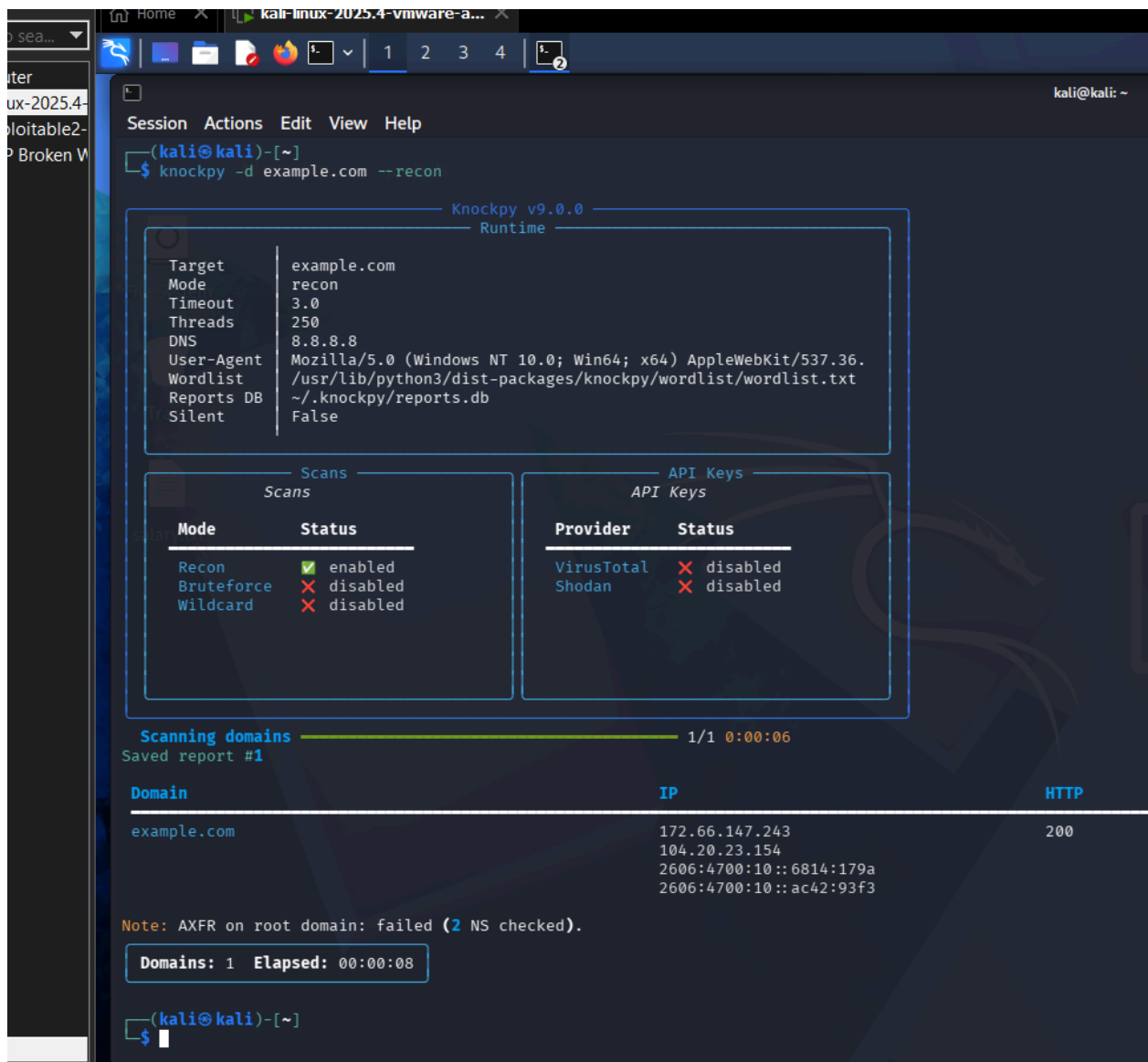
Exercise 3

Used KnockPy to enumerate subdomains for the following domain:

Target Domain: example.com

Commands used:

Knockpy -d example.com --recon



```
(kali@kali)-[~]
$ knockpy -d example.com --recon

Knockpy v9.0.0
Runtime
-----
Target      example.com
Mode        recon
Timeout     3.0
Threads     250
DNS         8.8.8.8
User-Agent  Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36.
Wordlist     /usr/lib/python3/dist-packages/knockpy/wordlist/wordlist.txt
Reports DB  ~/.knockpy/reports.db
Silent      False

Scans
-----
Mode      Status
Recon     ✓ enabled
Bruteforce ✗ disabled
Wildcard  ✗ disabled

API Keys
-----
Provider   Status
VirusTotal ✗ disabled
Shodan     ✗ disabled

Scanning domains 1/1 0:00:06
Saved report #1

Domain      IP                        HTTP
example.com 172.66.147.243
             104.20.23.154
             2606:4700:10::6814:179a
             2606:4700:10::ac42:93f3

Note: AXFR on root domain: failed (2 NS checked).

Domains: 1 Elapsed: 00:00:08

(kali@kali)-[~]
$
```

Findings

Discovered Subdomains and Information:

KnockPy checked subdomains for `example.com`

- It found no subdomains
- It tried DNS zone transfer (AXFR) but failed